Script - Controlled ACL – Restrict Record Access Based on Field Value

Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

Project Overview

Milestone-1 Creation of Users and Roles

Users Creation :

Log in to the ServiceNow instance with admin access.

Create a test user:

Navigate to User Administration → Users

Click New

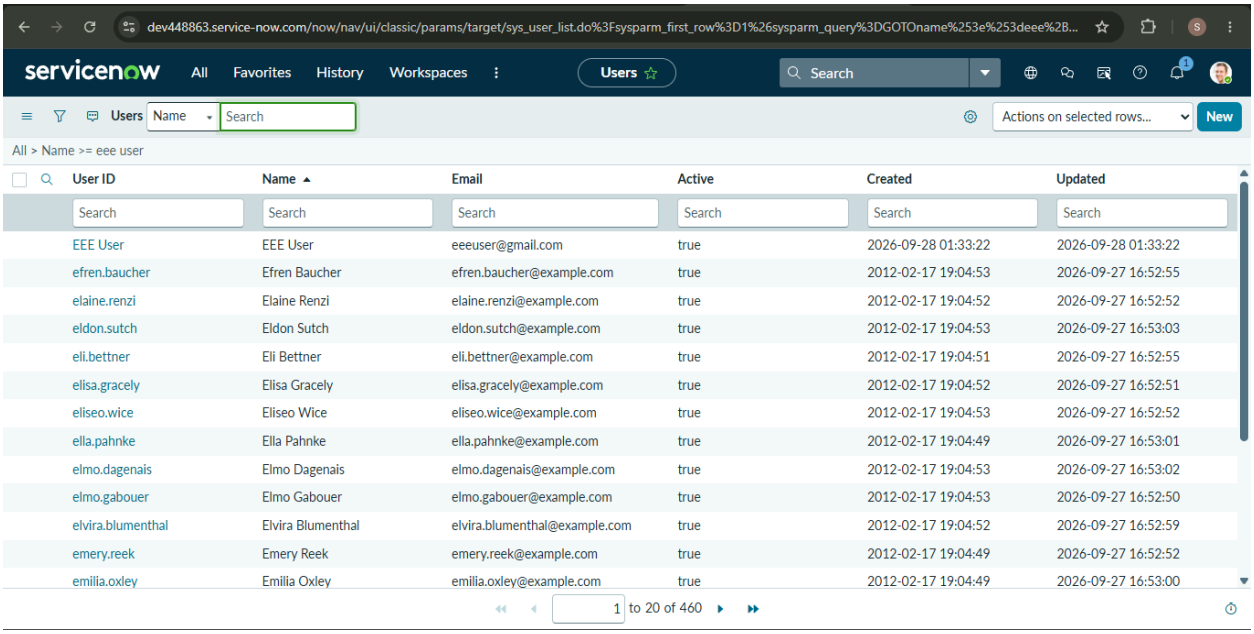
User ID: EEE User

First name: EEE

Last name: User

Email: eeeuser@gmail.com

Save the record



Roles Creation :

Navigate to User Administration → Roles

Click New

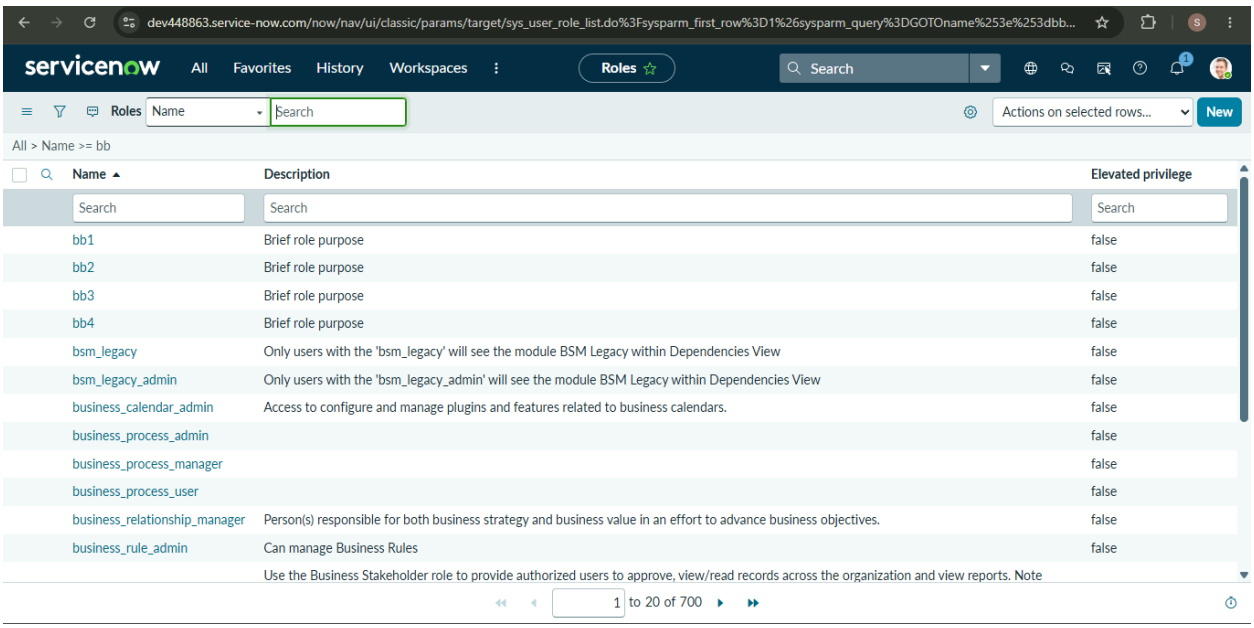
Name: Enter role name as bb1

Description: Brief role purpose

Click Submit

Create other roles also like bb2,bb3,bb4.(follow the same procedure)

Assign all the roles to the EEE User



Milestone-2 Tables Creation

In the Application Navigator, search for System Definition → Tables.

Click New and create a custom table with the following details

Label: Institution Details

Name: u_institution_details

Extends for : false

Save or Submit the form

In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture

The following fields to the u_institution_details table

Student Roll Number– Auto Number

Student Name – Reference- User

Faculty Name – Reference- User

Branch – Choice (ECE, EEE, CSE)

Email– String

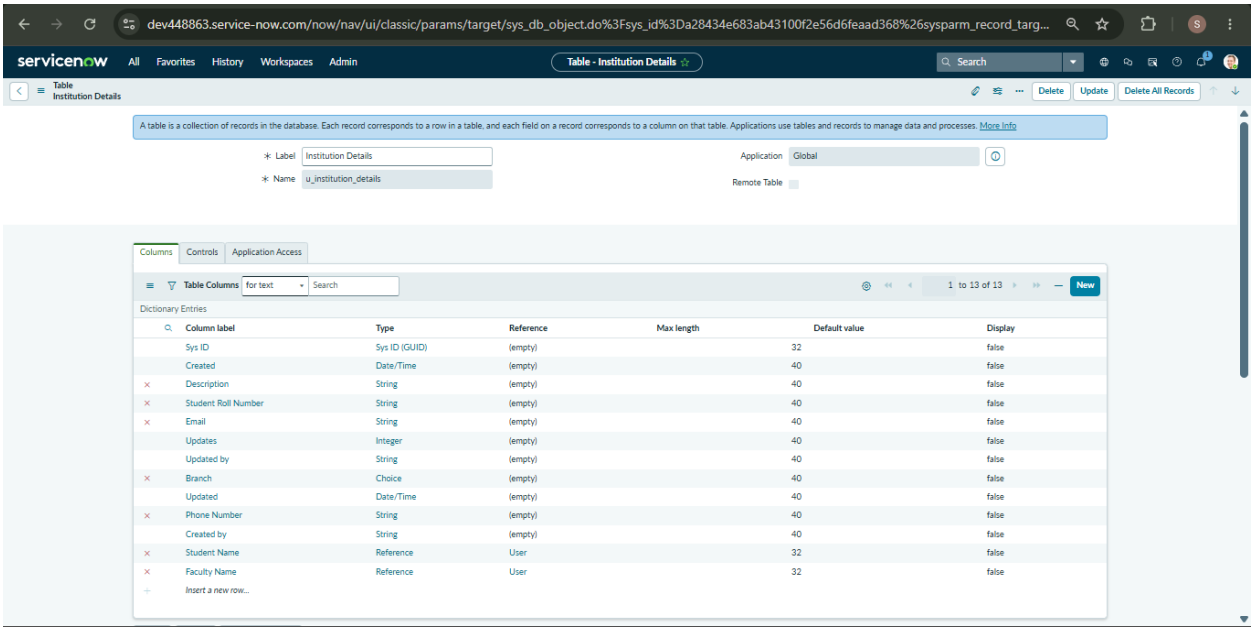
Phone Number– String

Description– Multi String

Save the Table

Click on the Show List in the Related Links

Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).



Milestone-3 Creations of Access Control List - READ

1.In the Application Navigator, search for System Security → Access Control (ACL).

2.Elevate “Security_admin” role

3.Click New to create a new ACL.

4.Enter the following details

Type: record

Operation: read

Name: u_institution_details

Active: true

Advanced: true

5.In the Requires role related list, add: bb1 (custom role)

6.Add Data condition as Branch is EEE

7.In the Script field, add the following script

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }
    // Allow only EEE branch users to see EEE records
    if (gs.hasRole('bb1')){
        return true;
    }
    // Deny access for all others
    return false;
})();
```

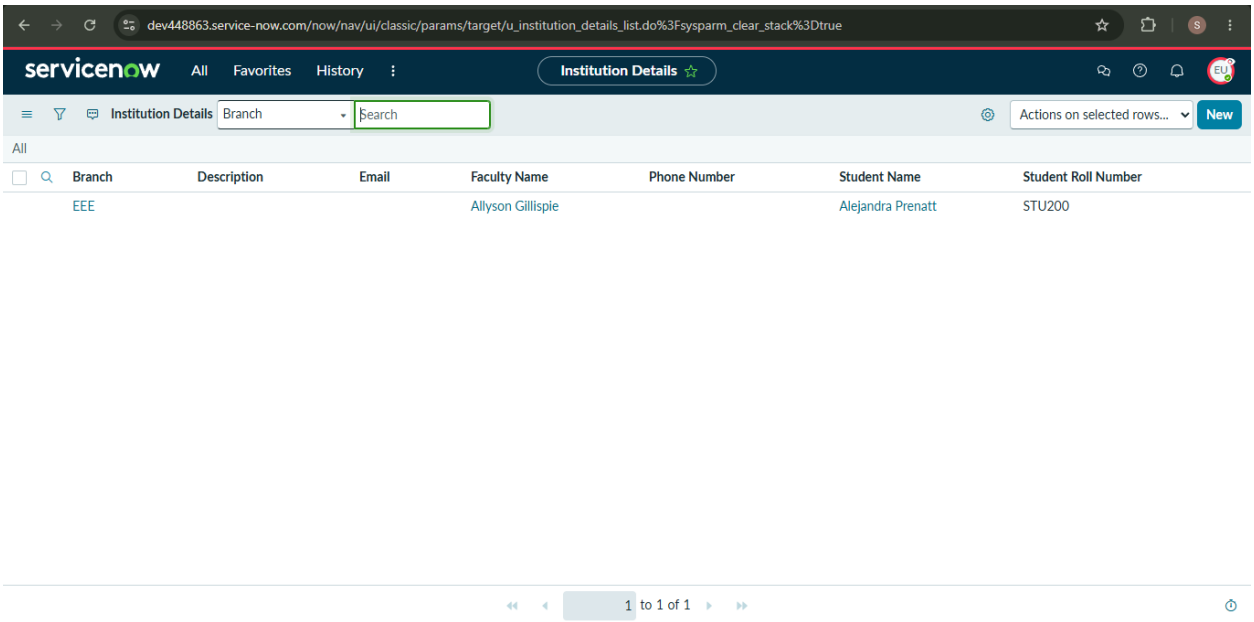
8.Click Submit to Save the ACL form

9.Impersonate EEE users and open the Student Records list.

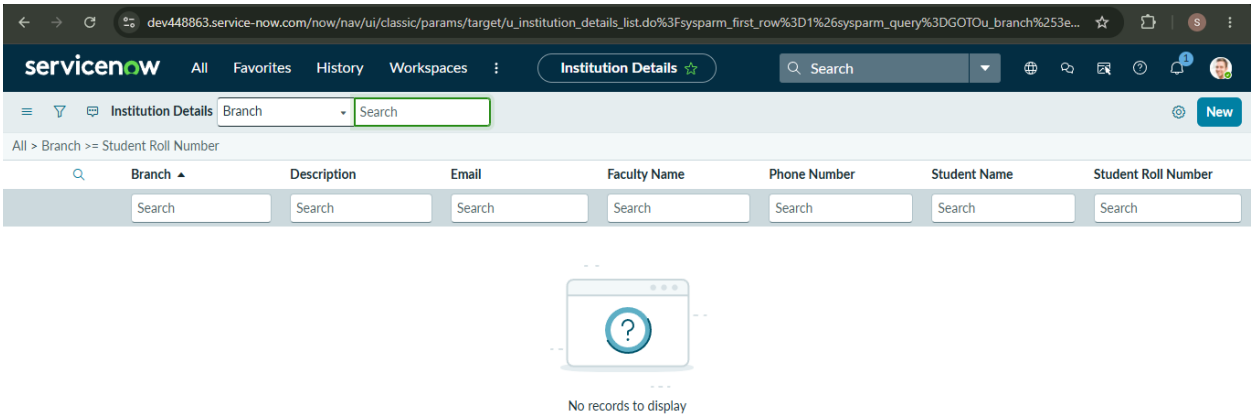
10.Copy the table name u_institution_details.list/LIST and paste in the All menu.

11.Impersonate Admin and open the Student Records list.

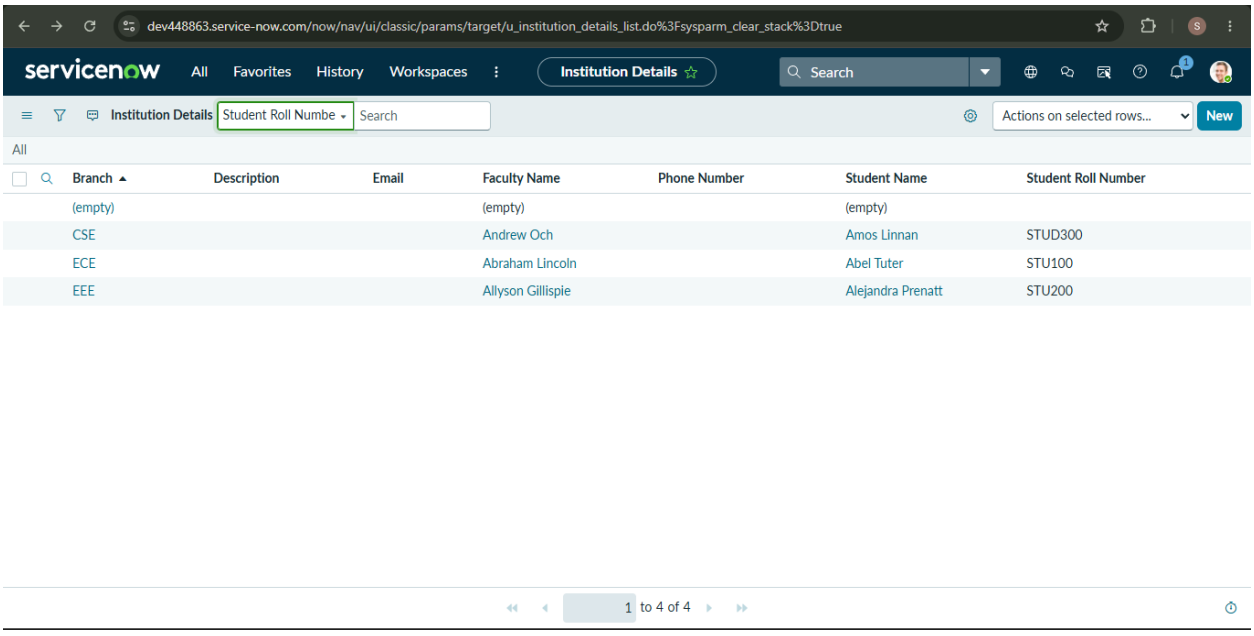
User with bb1 role can view only EEE branch records



User without the role cannot view any records



Admin user can view all records regardless of branch



Milestone-4 Creations of Access Control List - CREATE

1.Click New to create a new ACL(Create)

2.Enter the following details

Type: record

Operation: create

Name: u_institution_details

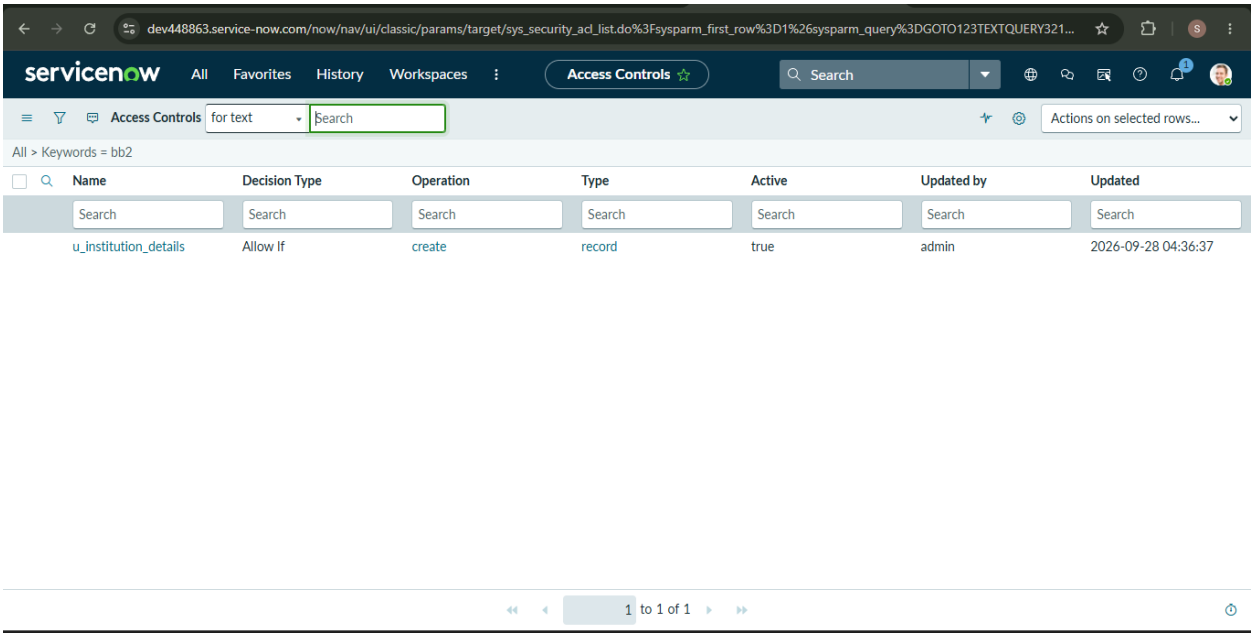
Active: true

3.In the Requires role related list, add: bb2 (custom role)

4.No need to add any Data condition

5.Save the ACL Record

User with bb1 &bb2 roles can view only EEE branch records & New Button



Milestone-5 Creations of Access Control List - WRITE

1.Click New to create a new ACL(Write)

2.Enter the following details

Type: record

Operation: Write

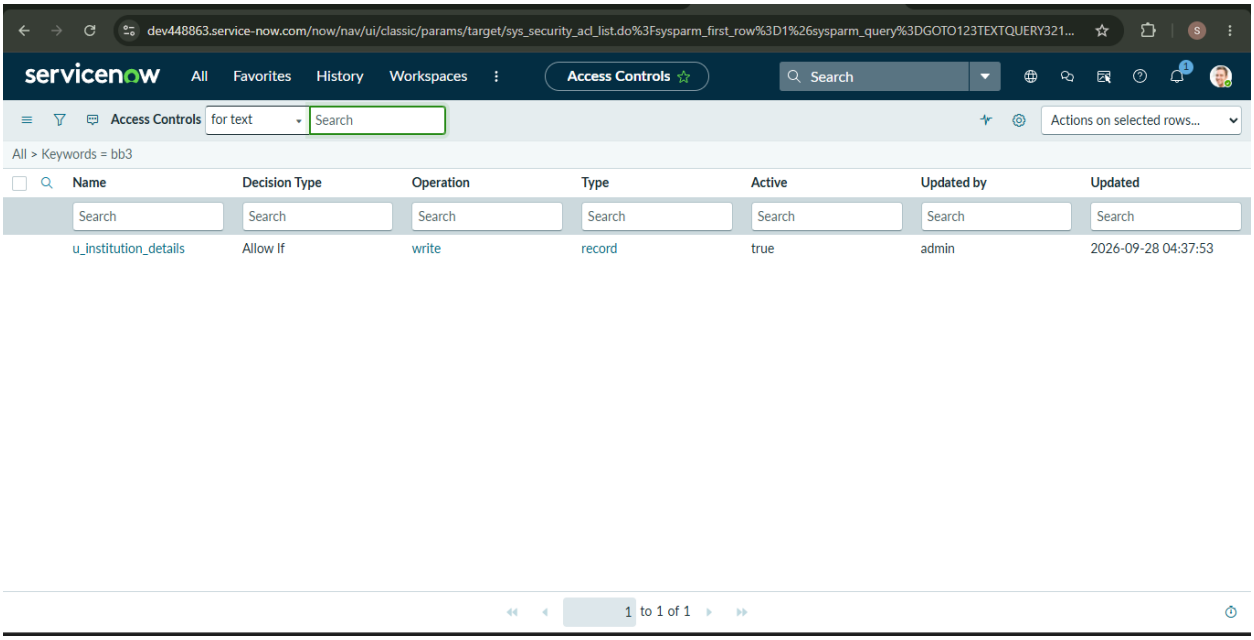
Name: u_institution_details

Active: true

3.In the Requires role related list, add: bb3 (custom role)

4.No need to add any Data condition

5.Save the ACL Record



Milestone-6 Creations of Access Control List - DELETE

1.Click New to create a new ACL(Delete)

2.Enter the following details

Type: record

Operation: Delete

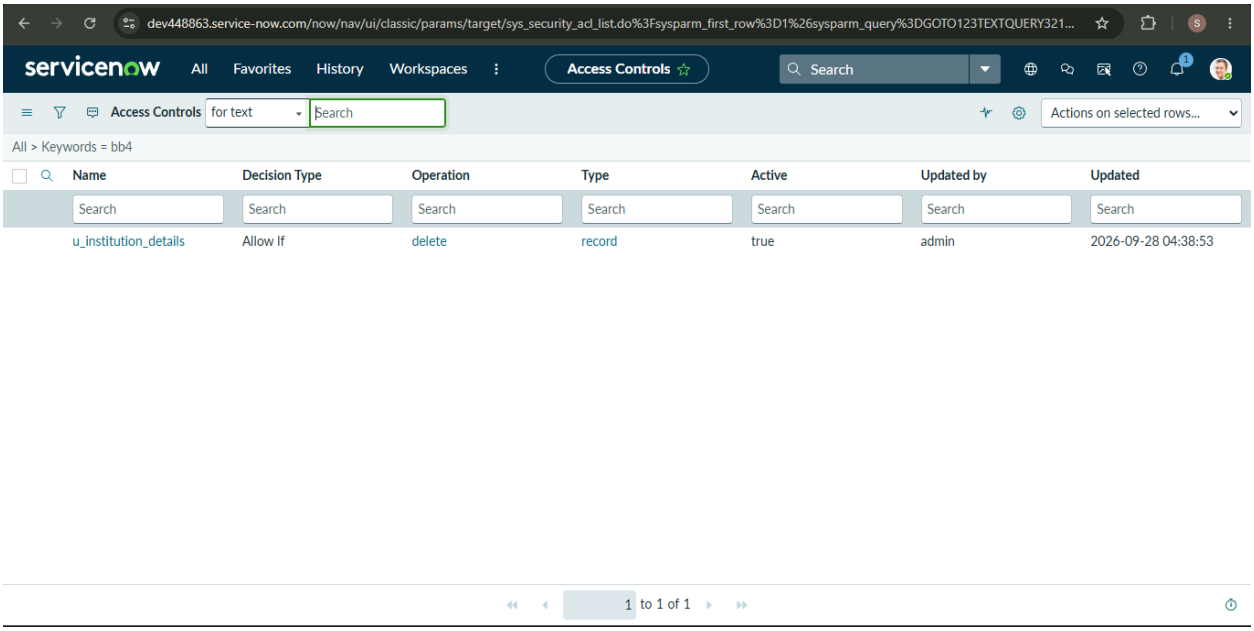
Name: u_institution_details

Active: true

3.In the Requires role related list, add: bb4 (custom role)

4.No need to add any Data condition

5.Save the ACL Record



Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.